

CHAPTER 9

PRESSURE

1.8 Pressure

- 1 Define pressure as force per unit area; recall and use the equation

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$p = \frac{F}{A}$$

- 2 Describe how pressure varies with force and area in the context of everyday examples
- 3 State that the pressure at a surface produces a force in a direction at right angles to the surface and describe an experiment to show this
- 4 Describe how the height of a liquid column in a liquid barometer may be used to determine the atmospheric pressure
- 5 Describe, quantitatively, how the pressure beneath the surface of a liquid changes with depth and density of the liquid
- 6 Recall and use the equation for the change in pressure beneath the surface of a liquid
change in pressure = density $\times$ gravitational field strength $\times$ change in height

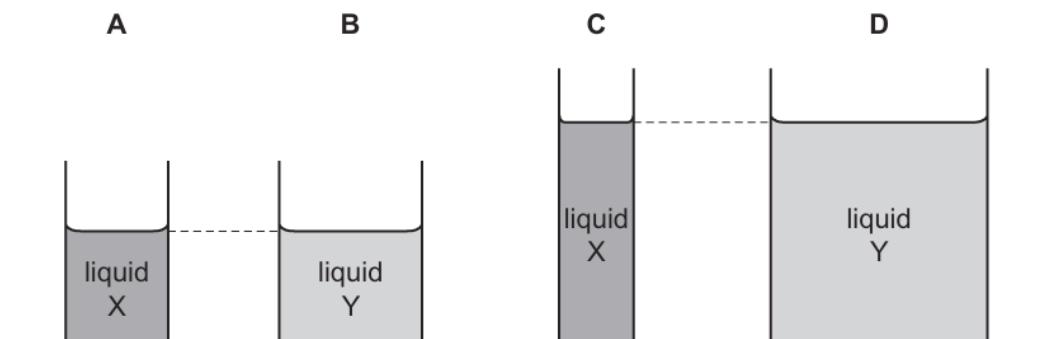
$$\Delta p = \rho g \Delta h$$

1. O/N/23/11/Q12

Two liquids, X and Y, are poured into beakers of different sizes.

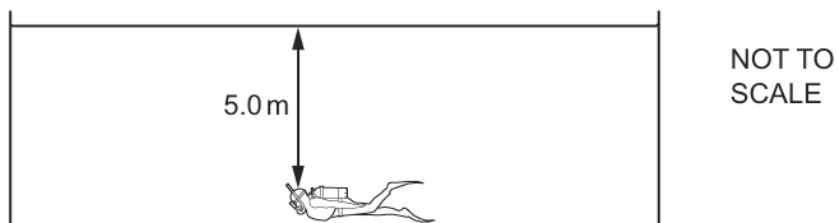
Liquid X has a density of 1000 kg/m^3 and liquid Y has a density of 900 kg/m^3 .

On the base of which beaker is the pressure greatest?



2. O/N/23/11/Q13

A swimmer is at the bottom of a pool. The density of the water in the pool is 1000 kg/m^3 and the gravitational field strength is 9.8 N/kg .



Which expression gives the pressure exerted on the swimmer due to the water?

- A $\frac{5.0 \times 1000}{9.8} \text{ Pa}$
- B $5.0 \times 1000 \times 9.8 \text{ Pa}$
- C $\frac{1000}{5.0} \text{ Pa}$
- D $\frac{1000 \times 9.8}{5.0} \text{ Pa}$

3. O/N/23/12/Q16

A man finds it difficult to hammer a wooden post into the ground.

How could he make the post go in more easily?

- A** Make the bottom of the post pointed.
- B** Make the top of the post pointed.
- C** Use a longer post.
- D** Use a wider post.

4. M/J/23/11/Q13

An African elephant has a mass of 6000 kg and each of its four feet has an area of 0.20 m^2 . The gravitational field strength g is 9.8 N/kg .

What is the average pressure due to the elephant when it is standing stationary on all four feet on level ground?

- A** 7.5 kPa **B** 30 kPa **C** 74 kPa **D** 290 kPa

5. O/N/22/11/Q13

Each tyre of a car has an area of 100 cm^2 in contact with the ground.

The car has a mass of 1600 kg. The weight of the car is equally distributed amongst the four tyres.

The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the ground?

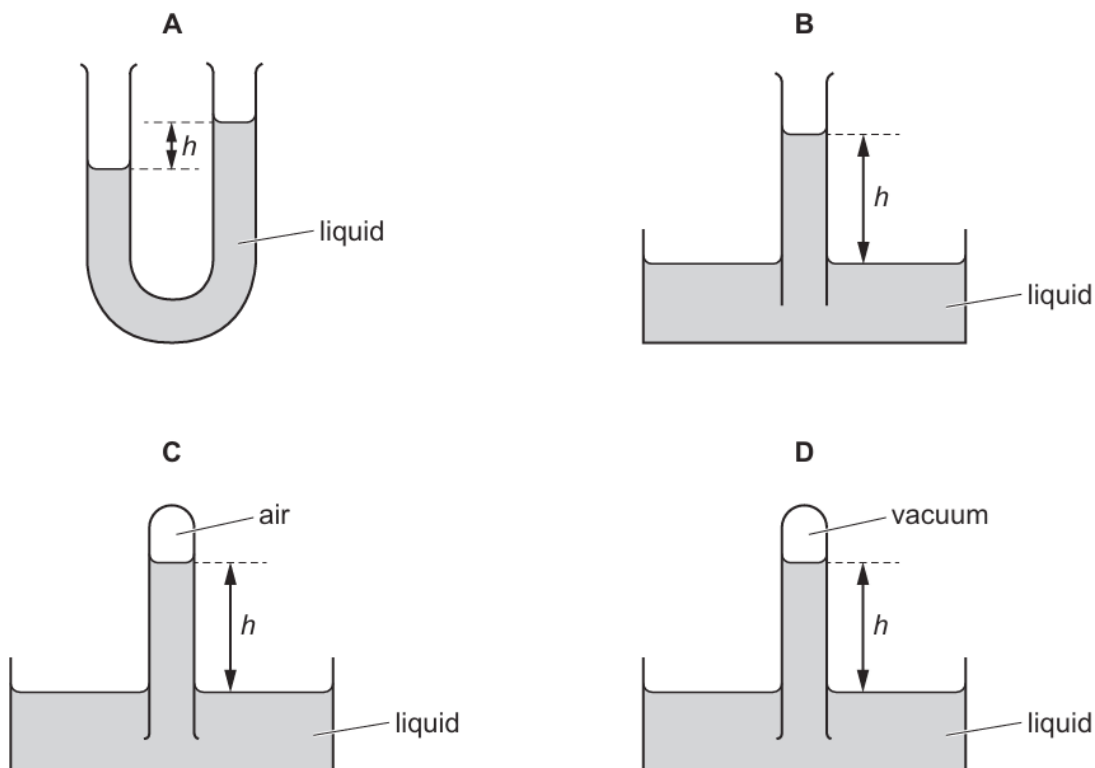
- A** 4.0 N/cm^2 **B** 16 N/cm^2 **C** 40 N/cm^2 **D** 160 N/cm^2

6. M/J/22/12/Q11

A barometer is an instrument used to measure atmospheric pressure.

In one type of barometer the height of a liquid in a tube is measured.

In which diagram does the height h represent atmospheric pressure?

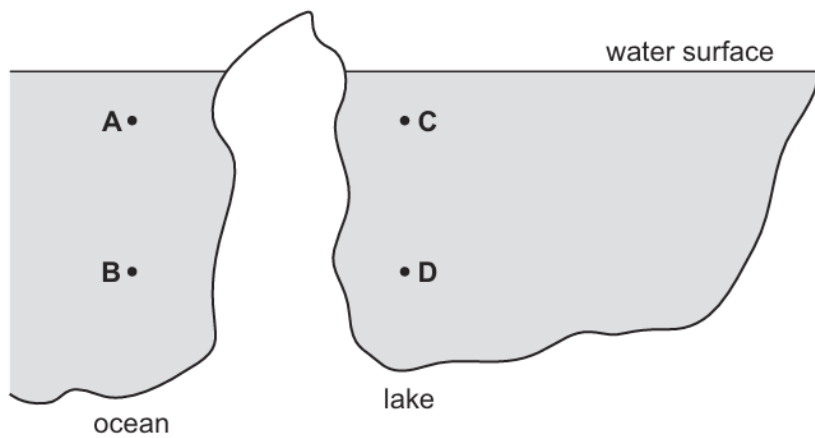


7. O/N/21/11/Q8

An underwater diver moves from the ocean to a fresh water lake.

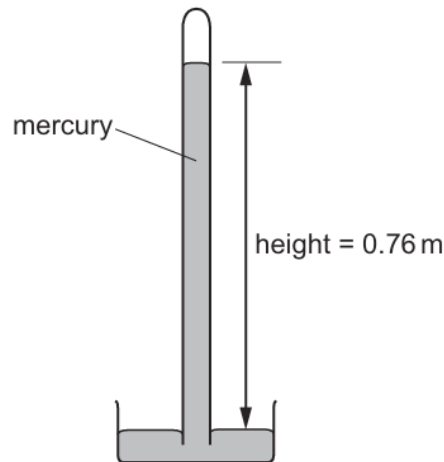
The density of water in the lake is less than in the ocean.

In which position does the diver experience the smallest pressure due to the water?



8. O/N/21/11/Q10

When the pressure exerted by the atmosphere is 0.10 MPa, the height of the mercury in a barometer is 0.76 m.



What is the pressure exerted by the atmosphere when the height of the mercury in the simple barometer is 0.57 m?

- A** 0.025 MPa **B** 0.075 MPa **C** 0.10 MPa **D** 0.13 MPa

9. M/J/21/11/Q11

Which expression for pressure is correct?

- A** force $\times$ area
B force $\div$ area
C mass $\times$ area
D mass $\div$ area

10. M/J/21/11/Q12

At a depth d in sea-water, the total pressure experienced by a diver is $2P$, where P is atmospheric pressure.

At which depth is the pressure $4P$?

- A** $1.5d$ **B** $2d$ **C** $3d$ **D** $4d$

11. M/J/21/12/Q12

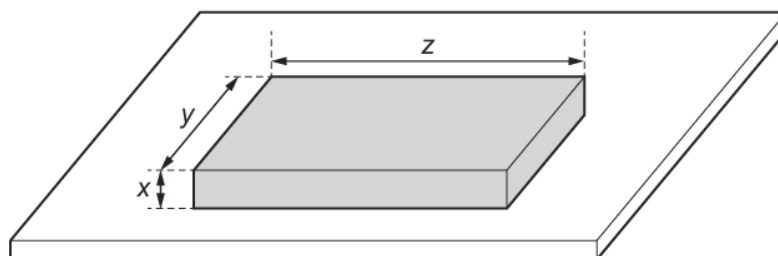
An object is placed at a depth d below the surface of a liquid of density ρ .

On which expression does the pressure on the object depend?

- A** $d + \rho$ **B** $d \times \rho$ **C** $d \div \rho$ **D** $\rho \div d$

12. M/J/21/12/Q13

The diagram shows the dimensions of a box of mass M and weight W at rest on a table.

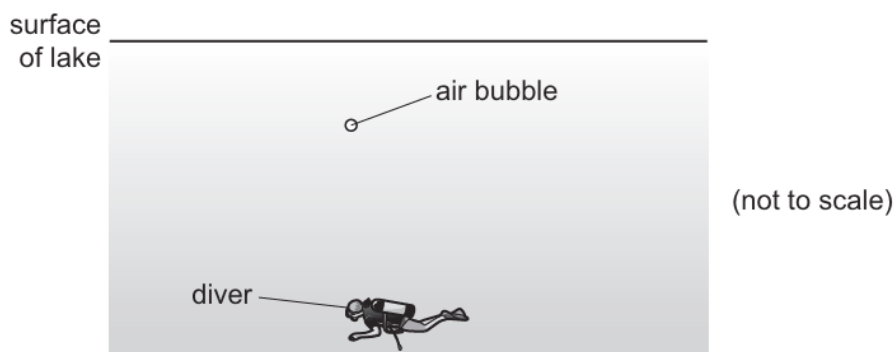


What is the pressure on the table due to the box?

- A** $\frac{M}{xyz}$ **B** $\frac{M}{yz}$ **C** $\frac{W}{xyz}$ **D** $\frac{W}{yz}$

13. O/N/20/12/Q12

The diagram shows a diver 20 m below the surface of a lake. The total pressure at this depth is 3.0×10^5 Pa.



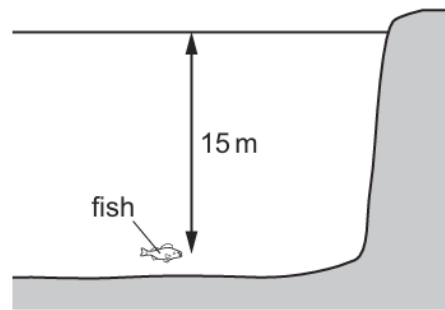
An air bubble has a volume of 0.60 cm^3 as it leaves the diver. It rises to the surface of the lake where the pressure is 1.0×10^5 Pa. The temperature of the air in the bubble remains constant.

What is the volume of the air bubble at the surface of the lake?

- A** 0.20 cm^3 **B** 0.60 cm^3 **C** 1.8 cm^3 **D** 2.4 cm^3

14. M/J/20/11/Q14

A fish is swimming 15 m below the surface of a lake, as shown.



The density of the water is 1000 kg/m^3 .

Atmospheric pressure is $100\,000 \text{ Pa}$.

The acceleration of free fall g is 10 m/s^2 .

What is the total pressure on the fish?

- A** $50\,000 \text{ Pa}$ **B** $120\,000 \text{ Pa}$ **C** $150\,000 \text{ Pa}$ **D** $250\,000 \text{ Pa}$

15. M/J/20/12/Q14

An object is placed at different depths in liquids of different densities.

In which liquid and at what depth is the pressure on the object the greatest?

	<u>density of liquid</u> kg/m^3	depth / m
A	1000	4.0
B	1000	10
C	1200	4.0
D	1200	10

16. O/N/19/11/Q15

The list gives the symbols for some physical quantities.

acceleration of free-fall g

atmospheric pressure P

density of water ρ

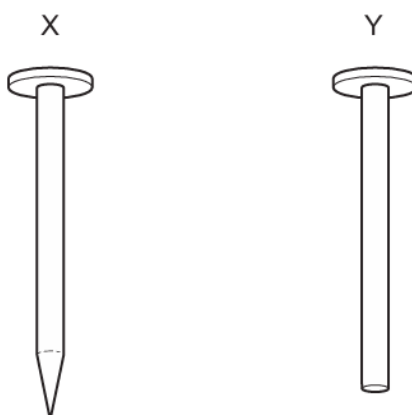
depth of water h

Which quantities from the list are used to calculate the total pressure at the bottom of a lake?

- A** g , h , P and ρ
B g , h and P only
C h , P and ρ only
D g , h and ρ only

17. O/N/19/11/Q14

Nail X is tapped into wood. Nail Y is tapped into the same wood using the same force.

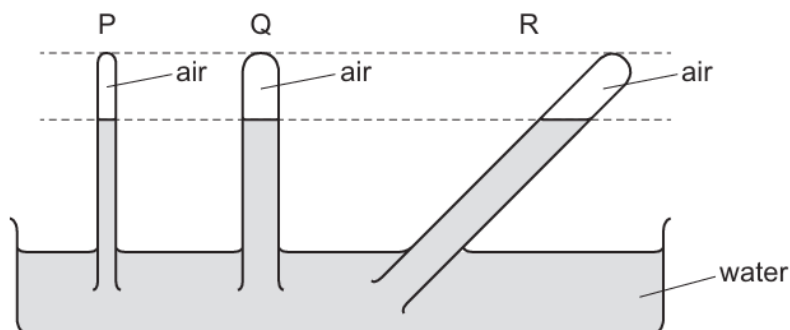


Which nail moves furthest into the wood and why?

	nail	why
A	X	X produces greater pressure on the wood
B	X	X produces smaller pressure on the wood
C	Y	Y produces greater pressure on the wood
D	Y	Y produces smaller pressure on the wood

18. O/N/19/12/Q12

The diagram shows three tubes P, Q and R. Each tube contains air trapped by a water column.



Which statement is correct?

- A** The pressure of the trapped air is equal in tubes P, Q and R.
- B** The pressure of the trapped air is greatest in tube Q.
- C** The pressure of the trapped air is greatest in tube R.
- D** The pressure of the trapped air is greatest in tube P.

19. M/J/19/11/Q11

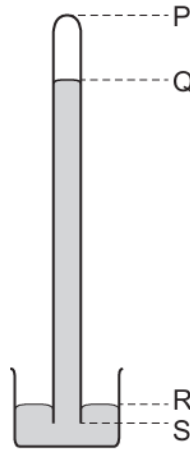
A garden table weighs 60 N and has a top surface of area 2.0 m^2 . It is raining and the rain produces a pressure of 4.0 N/m^2 on the table.

Ignoring the pressure of the atmosphere, what is the force exerted by the table on the ground?

- A** 52 N **B** 58 N **C** 62 N **D** 68 N

20. O/N/18/11/Q11

A long tube, full of mercury, is inverted in a small dish of mercury.



The mercury level in the tube falls, leaving a vacuum at the top.

When the atmospheric pressure decreases, which length decreases?

- A** PQ **B** PS **C** QR **D** RS

21. O/N/18/12/Q13

Different liquids are poured into four different containers.

In which container does the liquid produce the greatest pressure at the bottom of the container?

	area of base / cm^2	density of liquid / g/cm^3	depth of liquid / cm
A	10	1.3	50
B	20	0.80	80
C	40	1.0	60
D	50	0.92	75

22. O/N/18/12/Q14

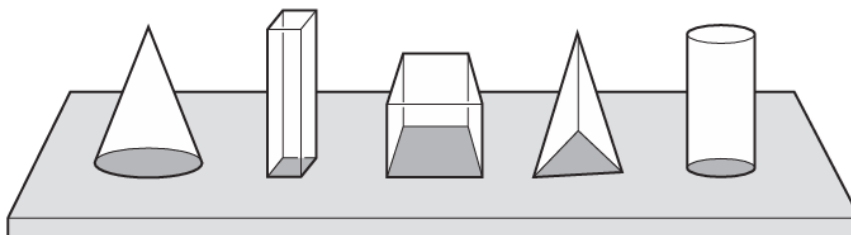
A rectangular block of metal has weight 6.0 N and measures $3.0\text{ cm} \times 4.0\text{ cm} \times 5.0\text{ cm}$.

What is the smallest pressure that it can exert when resting on a horizontal surface?

- A** 0.10 N/cm^2 **B** 0.30 N/cm^2 **C** 0.40 N/cm^2 **D** 0.50 N/cm^2

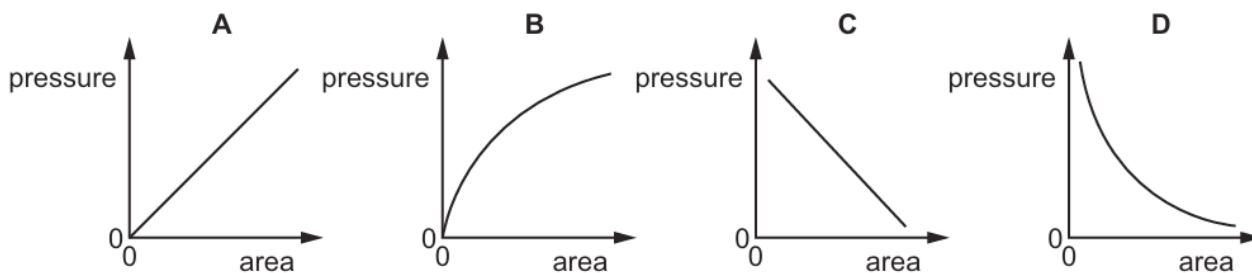
23. M/J/18/11/Q14

Five blocks have the same mass but different base areas. They all rest on a horizontal table.



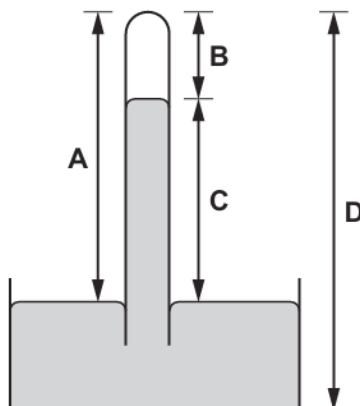
A graph is plotted to show the relationship between the pressure exerted on the table and the base area of the block.

Which graph shows this relationship?

**24. M/J/18/11/Q16**

The diagram shows a simple mercury barometer.

Which height is a measure of the atmospheric pressure?



25. M/J/18/12/Q18

A sealed packet containing air and a snack is purchased at an airport. The sealed packet is taken on board an aircraft. During the flight the packet becomes larger.

What causes the packet to become larger?

- A** The density of the air inside the packet increases.
- B** The mass of the packet increases.
- C** The pressure of the air outside the packet decreases.
- D** The volume of the air inside the packet decreases.

26. M/J/18/12/Q19

A sealed container of gas is heated and the pressure inside increases.

What happens to the molecules of the gas to cause this increase in pressure?

- A** Their kinetic energy decreases.
- B** They become heavier.
- C** They expand.
- D** They hit the container more frequently.

27. O/N/17/12/Q12

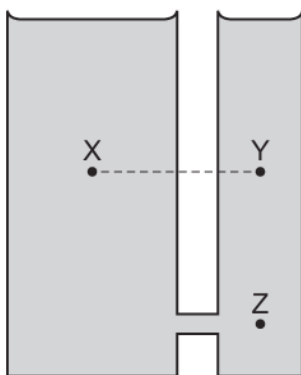
An extremely large increase in pressure is needed to compress a liquid. A gas can be compressed by a much smaller increase in pressure.

Which statement explains this?

- A** Molecules repel each other very strongly when very close.
- B** The attractive forces between molecules are small at large distances.
- C** The molecules in the gas collide with the walls of their container and rebound.
- D** The molecules of a liquid are constantly moving at random.

28. M/J/17/11/Q16

Two cylindrical vessels are joined together and filled with water as shown.



How does the pressure at point X compare to the pressure at points Y and Z?

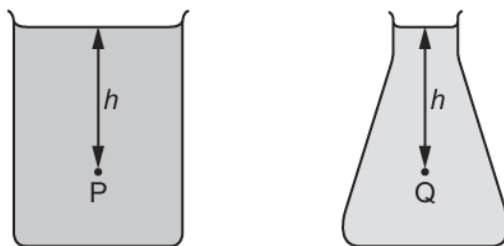
	compared to Y	compared to Z
A	pressure at X is higher	pressure at X is lower
B	pressure at X is higher	pressure at X is the same
C	pressure at X is the same	pressure at X is lower
D	pressure at X is the same	pressure at X is the same

29. M/J/17/12/Q18

Two glass containers filled with different liquids are placed next to each other.

Point P is a distance h below the surface of the liquid in one container.

Point Q is a distance h below the surface of the liquid in the other container.



Why is the pressure at P different from the pressure at Q?

- A** The atmospheric pressure is different at P.
- B** The densities of the liquids are different.
- C** The gravitational field strength is different at P.
- D** The shapes of the containers are different.

30. O/N/16/11/Q12

The pressure of the atmosphere is 100 000 Pa.

What force does the atmosphere exert on the upper surface of a pond of surface area 20 m²?

- A** 5000 N **B** 100 000 N **C** 100 020 N **D** 2 000 000 N

31. O/N/16/11/Q13

Oil of density $8.5 \times 10^2 \text{ kg/m}^3$ is stored in a large tank.

The gravitational field strength g is 10 N/kg.

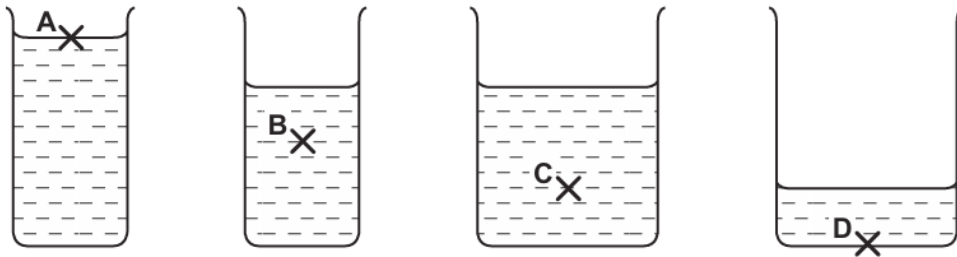
What is the pressure due to the oil 6.0 m below its surface?

- A** 51 Pa **B** 510 Pa **C** 5100 Pa **D** 51 000 Pa

32. M/J/16/11/Q15

Four beakers contain the same liquid.

At which point is the pressure the greatest?



33. M/J/16/11/Q16

Water of depth 10 m exerts a pressure equal to atmospheric pressure.

An air bubble rises to the surface of a lake which is 20 m deep. When the bubble reaches the surface, its volume is 6.0 cm³.

What is the volume of the air bubble at the bottom of the lake?

- A** 2.0 cm³ **B** 3.0 cm³ **C** 12 cm³ **D** 18 cm³

34. M/J/16/12/Q11

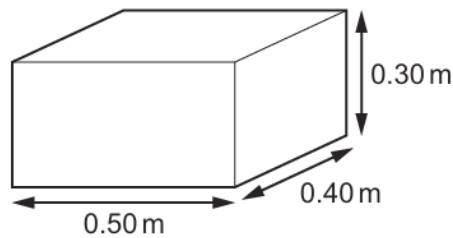
A block of weight W rests on a side of area A . The gravitational field strength is g .

What is the pressure exerted on the ground due to the block?

- A** WA **B** $\frac{W}{A}$ **C** $\frac{WA}{g}$ **D** $\frac{W}{g}$

35. O/N/15/11/Q11

A block of weight 900 N has rectangular faces. The diagram shows the lengths of the sides.



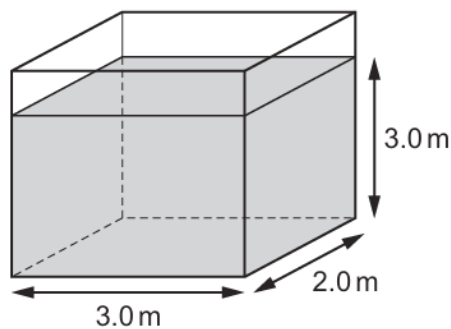
The block can rest on any of its faces.

What is the minimum pressure that the block can exert on the ground when resting on one of its faces?

- A** 900 Pa **B** 4500 Pa **C** 6000 Pa **D** 7500 Pa

36. O/N/15/12/Q13

The base of a rectangular storage tank is 2.0 m by 3.0 m. The tank is filled with paraffin to a depth of 3.0 m.



The density of paraffin is 800 kg/m^3 and the gravitational field strength is 10 N/kg .

What is the pressure at the bottom of the tank due to the paraffin?

- A** 2400 Pa **B** 14 400 Pa **C** 24 000 Pa **D** 144 000 Pa

37. O/N/15/12/Q15

A piston of area 10 cm^2 is pushed slowly into a very large cylinder containing gas at a pressure of 10 N/cm^2 . The pressure of the gas remains constant as the piston moves a distance of 0.10 m.

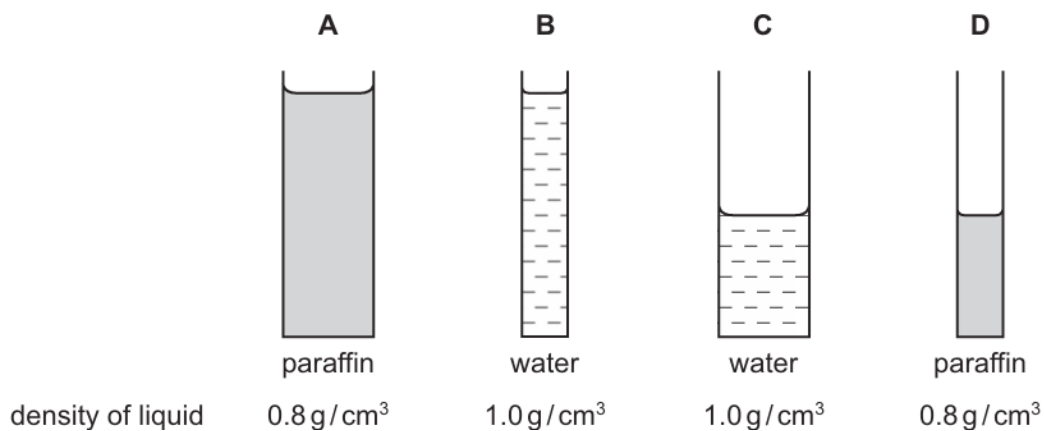
What is the force of the gas on the piston and what is the work done by the piston on the gas?

	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

38. M/J/15/11/Q14

The diagrams show liquids in containers.

Which column of liquid exerts the greatest pressure on the base of its container?



39. O/N/14/11/Q10

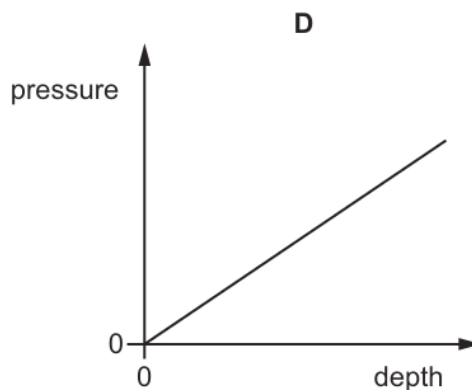
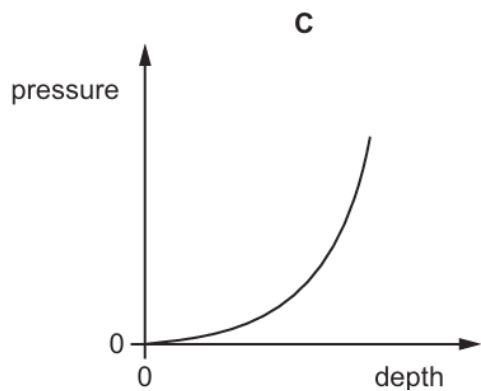
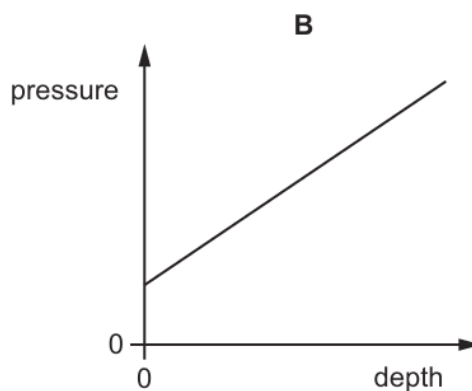
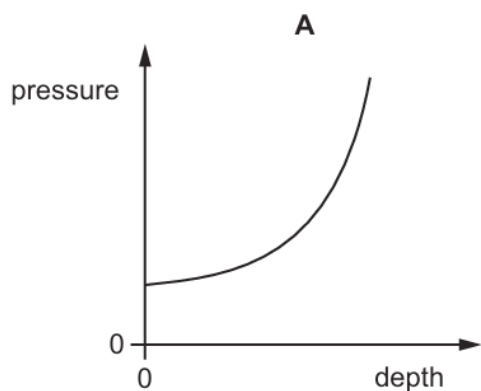
Objects of different weights are placed on a rigid, horizontal surface.

Which row shows the correct pressure acting on the surface?

	weight / N	area in contact / m^2	pressure / Pa
A	10	0.1	1
B	20	0.2	0.01
C	30	0.1	300
D	40	0.2	8

40. M/J/14/12/Q14

Which graph shows the total external pressure acting on a submarine at different depths below the surface of the sea?



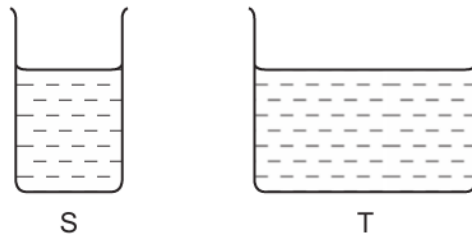
41. O/N/13/11/Q9

Which source releases carbon dioxide, a greenhouse gas, when generating electricity?

- A** fossil fuels
- B** geothermal
- C** hydroelectric
- D** nuclear

42. O/N/13/12/Q10

Two vessels S and T are filled to the same level with the same liquid. The area of the base of S is less than that of T.

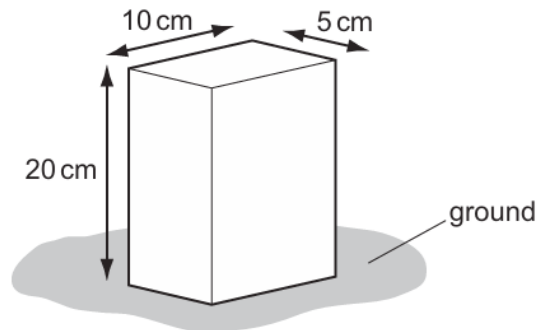


Which statement is correct?

- A** The force on the base of S is greater than the force on the base of T.
- B** The force on the base of S is the same as the force on the base of T.
- C** The pressure on the base of S is greater than the pressure on the base of T.
- D** The pressure on the base of S is the same as the pressure on the base of T.

43. M/J/13/11/Q8

A brick of weight 80 N stands upright on the ground as shown.



What is the pressure it exerts on the ground?

- A** 0.080 N/cm^2 **B** 0.40 N/cm^2 **C** 0.80 N/cm^2 **D** 1.6 N/cm^2

44. O/N/12/11/Q11

A garden table weighs 40 N and has a top surface of area 2 m^2 . It is raining and the rain produces a pressure of 4 N/m^2 on the table.

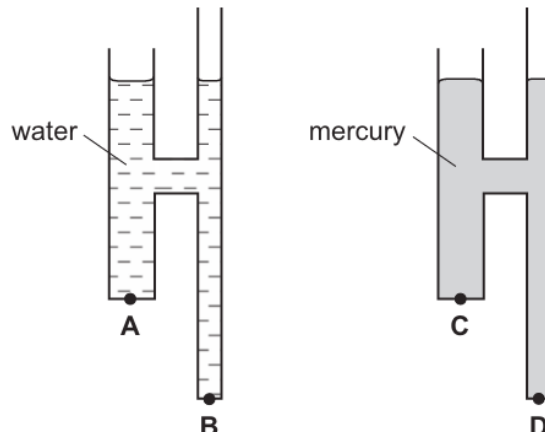
What is the force exerted by the table on the ground?

- A** 20 N **B** 32 N **C** 42 N **D** 48 N

45. O/N/11/11/Q13

The diagram shows two identical pieces of apparatus. One is filled with water and the other is filled with mercury. Water is less dense than mercury.

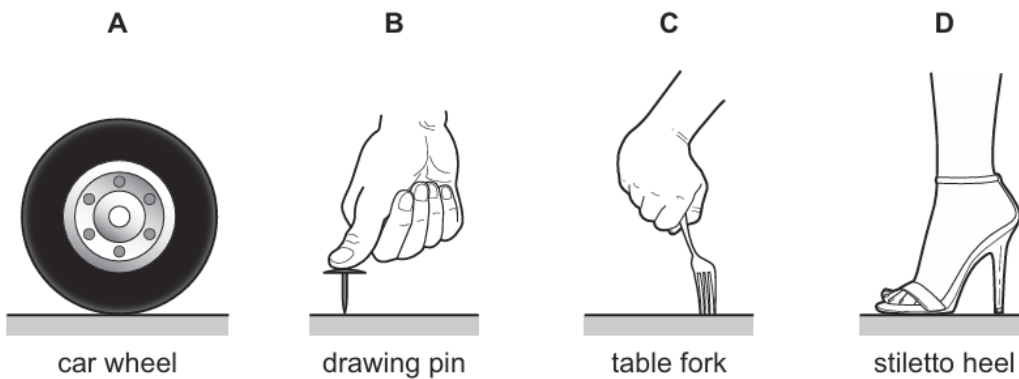
At which point is the liquid pressure greatest?



46. M/J/11/11/Q10

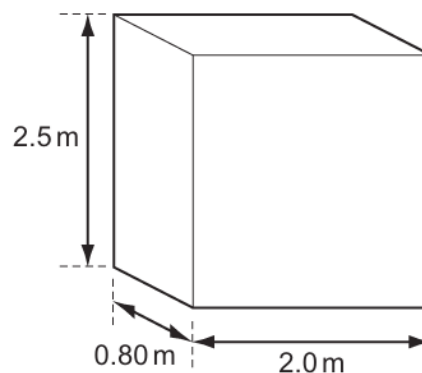
The **same** downward force is applied to four objects resting on a horizontal surface.

Which exerts the greatest pressure on the surface?



47. O/N/10/11/Q11

The base for a statue rests on level ground. It is made from stone and is 2.0 m long, 2.5 m high and 0.80 m wide. It has a weight of 96 000 N.

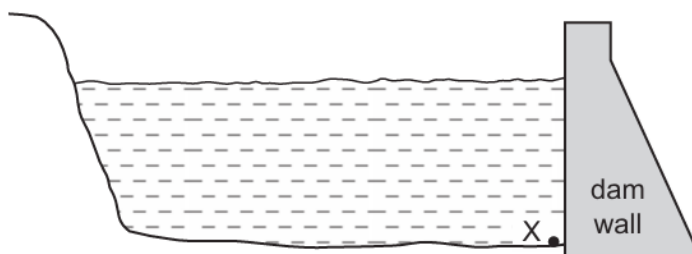


What is the pressure that the base exerts on the ground?

- A** 19 kPa **B** 24 kPa **C** 48 kPa **D** 60 kPa

48. O/N/10/11/Q12

An engineer designs a dam wall for a reservoir.

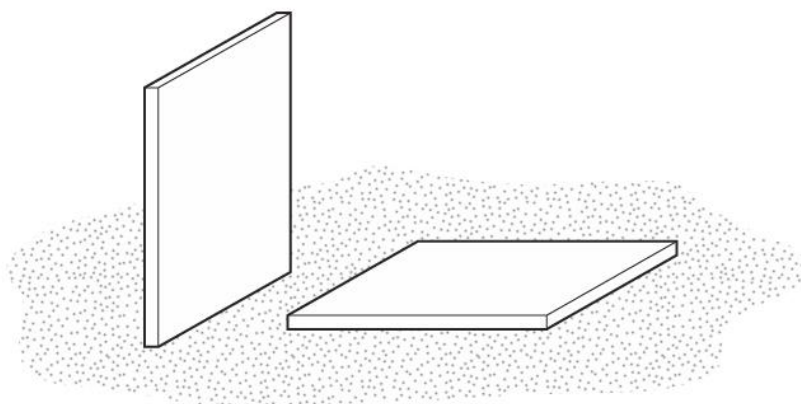


Which factor determines the pressure at X?

- A** the depth of the water in the reservoir
- B** the surface area of the reservoir
- C** the length of the reservoir
- D** the thickness of the dam wall

49. M/J/09/Q10

A builder leaves two identical, heavy, stone tiles resting on soft earth. One is vertical and the other is horizontal.



After a few hours, the vertical tile has started to sink into the soft earth, but the horizontal one has not.

Which row correctly compares the forces and the pressures that the tiles exert on the earth?

	forces	pressures
A	different	different
B	different	same
C	same	different
D	same	same

50. M/J/08/Q12

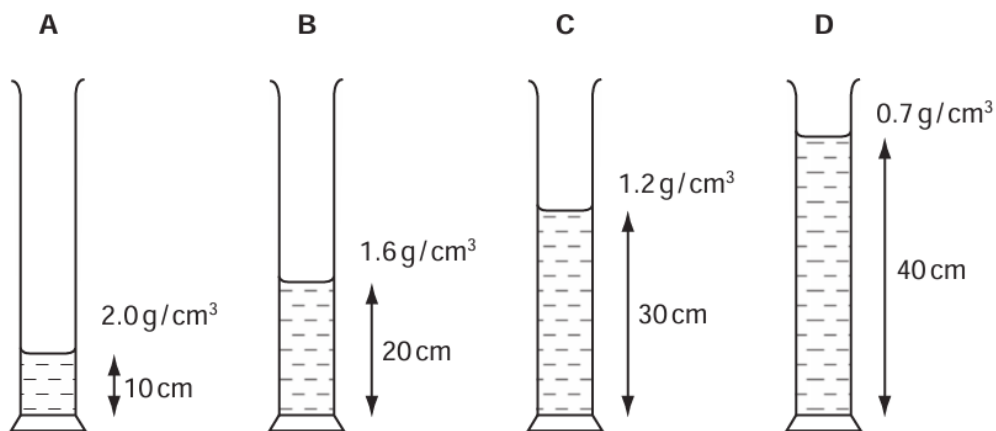
Which statement about the pressure in a column of liquid is correct?

- A** It acts only vertically downwards.
- B** It increases if the column width increases.
- C** It increases with depth in the column.
- D** It is uniform throughout the column.

51. O/N/07/Q12

Four different liquids are poured into identical measuring cylinders. The diagrams show the depths of the liquids and their densities.

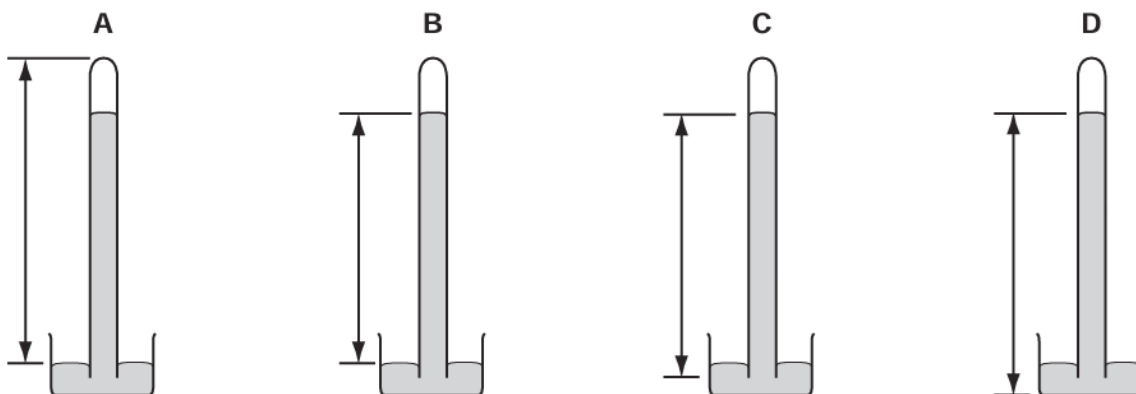
Which liquid causes the largest pressure on the base of its measuring cylinder?



52. O/N/06/Q10

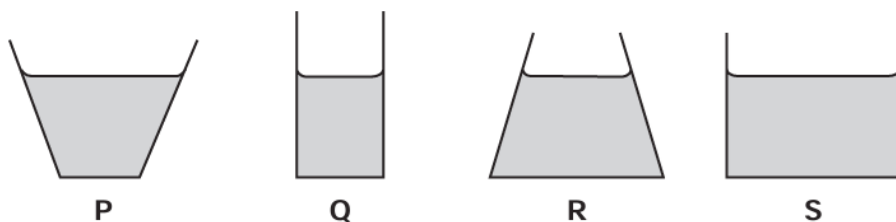
The diagrams show a simple mercury barometer.

Which diagram shows the distance to be measured to find atmospheric pressure?



53. M/J/06/Q14

The diagrams show, to the same scale, the vertical sections of a set of circular vessels. Each vessel contains the same depth of water.

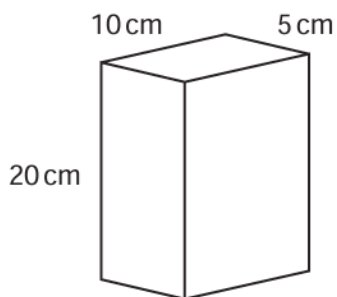


Which of the following statements is correct?

- A** The water exerts the greatest pressure on the base of vessel **P**.
- B** The water exerts the greatest pressure on the base of vessel **S**.
- C** The water exerts the same force on the base of each vessel.
- D** The water exerts the same pressure on the base of each vessel.

54. O/N/05/Q13

A brick of weight 80 N stands upright on the ground as shown.



What is the pressure it exerts on the ground?

- A** $\frac{80}{20 \times 10} \text{ N/cm}^2$
- B** $\frac{20 \times 10}{80} \text{ N/cm}^2$
- C** $\frac{80}{10 \times 5} \text{ N/cm}^2$
- D** $\frac{10 \times 5}{80} \text{ N/cm}^2$

55. M/J/05/Q12

What does **not** affect the pressure at a point beneath the surface of a liquid?

- A area of the liquid surface
- B density of the liquid
- C depth of the point below the surface
- D strength of the gravitational field

56. M/J/05/Q13

A small table weighing 40 N stands on four legs, each having an area of 0.001 m².

What is the pressure of the table on the floor?

- A 400 N/m² B 1000 N/m² C 10 000 N/m² D 40 000 N/m²

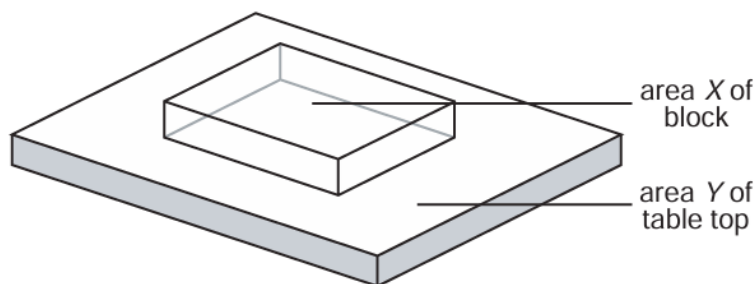
57. O/N/04/Q14

Which would be the **least** likely to sink into soft ground?

- A a loaded lorry with four wheels
- B a loaded lorry with six wheels
- C an empty lorry with four wheels
- D an empty lorry with six wheels

58. M/J/04/Q14

The diagram shows a glass block resting on a table top.



The area of the block in contact with the table is X and the area of the table top is Y .

The weight of the block is P and the weight of the table is Q .

Which expression gives the pressure exerted on the table by the block?

- A $\frac{P}{X}$ B $\frac{P}{Y}$ C $\frac{Q}{X}$ D $\frac{Q}{Y}$

ANSWERS

- | | |
|-------|-------|
| 1. C | 48. A |
| 2. B | 49. C |
| 3. A | 50. C |
| 4. C | 51. C |
| 5. C | 52. B |
| 6. D | 53. D |
| 7. C | 54. C |
| 8. B | 55. A |
| 9. B | 56. C |
| 10. C | 57. D |
| 11. B | 58. A |
| 12. D | |
| 13. C | |
| 14. D | |
| 15. D | |
| 16. A | |
| 17. A | |
| 18. A | |
| 19. D | |
| 20. C | |
| 21. D | |
| 22. B | |
| 23. D | |
| 24. C | |
| 25. C | |
| 26. D | |
| 27. A | |
| 28. C | |
| 29. B | |
| 30. D | |
| 31. D | |
| 32. C | |
| 33. A | |
| 34. B | |
| 35. B | |
| 36. C | |
| 37. C | |
| 38. B | |
| 39. C | |
| 40. B | |
| 41. C | |
| 42. D | |
| 43. D | |
| 44. D | |
| 45. D | |
| 46. B | |
| 47. D | |

1. O/N/23/21/Q1

Fig. 1.1 shows a toy water rocket.

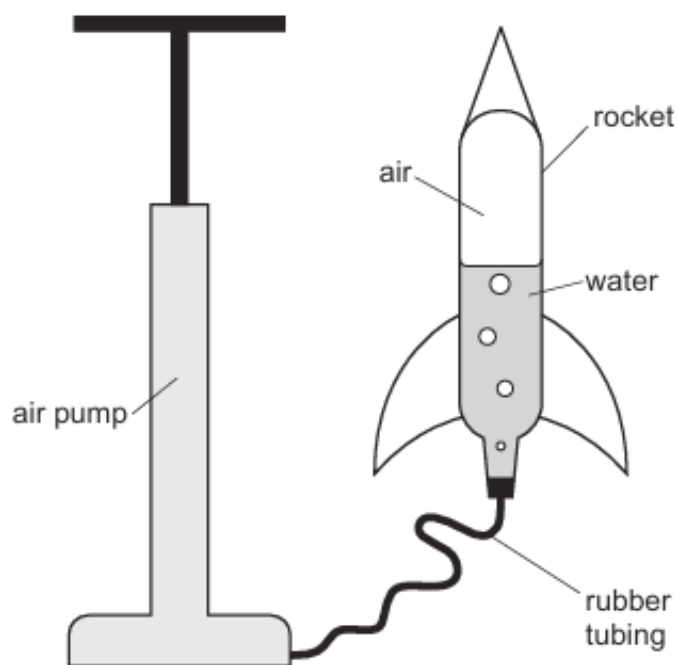


Fig. 1.1

The rocket is half-filled with water and connected by rubber tubing to a pump.

(a) Air is then pumped into the rocket so that the pressure of the air inside the rocket increases.

- (i) Explain, in terms of particles, why the air in the rocket exerts a pressure on the walls of the rocket.

.....
.....
.....
..... [3]

- (ii) The temperature of the air in the rocket remains constant as air is pumped into the rocket.

Explain why increasing the number of air particles in the rocket increases the pressure of the air in the rocket.

.....
..... [1]

2. O/N/23/22/Q4

Fig. 4.1 shows a long vertical tube, sealed at the base and open at the other end.

The tube is 1.0 m long.

The cross-sectional area of the tube is $4.0 \times 10^{-4} \text{ m}^2$.

It is filled with a liquid of density $1.4 \times 10^4 \text{ kg/m}^3$.

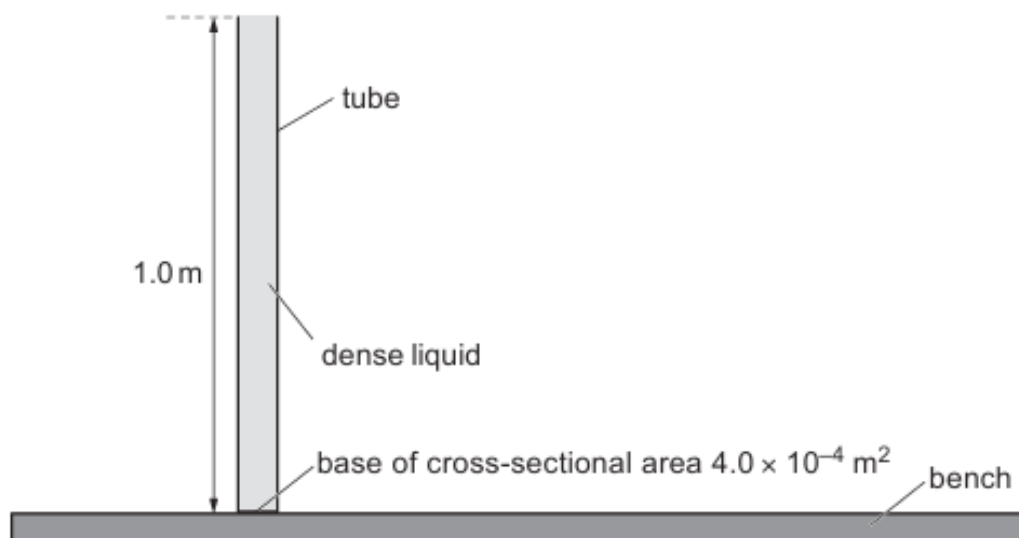


Fig. 4.1 (not to scale)

The atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

(a) Calculate:

(i) the total pressure in the liquid at the bottom of the tube

total pressure = Pa [3]

(ii) the force exerted on the inside surface of the bottom of the tube.

force = N [2]

- (b) A small sheet of glass is placed over the open end of the tube.

The tube is inverted in a container of the dense liquid.

The open end of the tube and the sheet of glass are a short distance below the surface of the liquid.

Fig. 4.2 shows the arrangement.

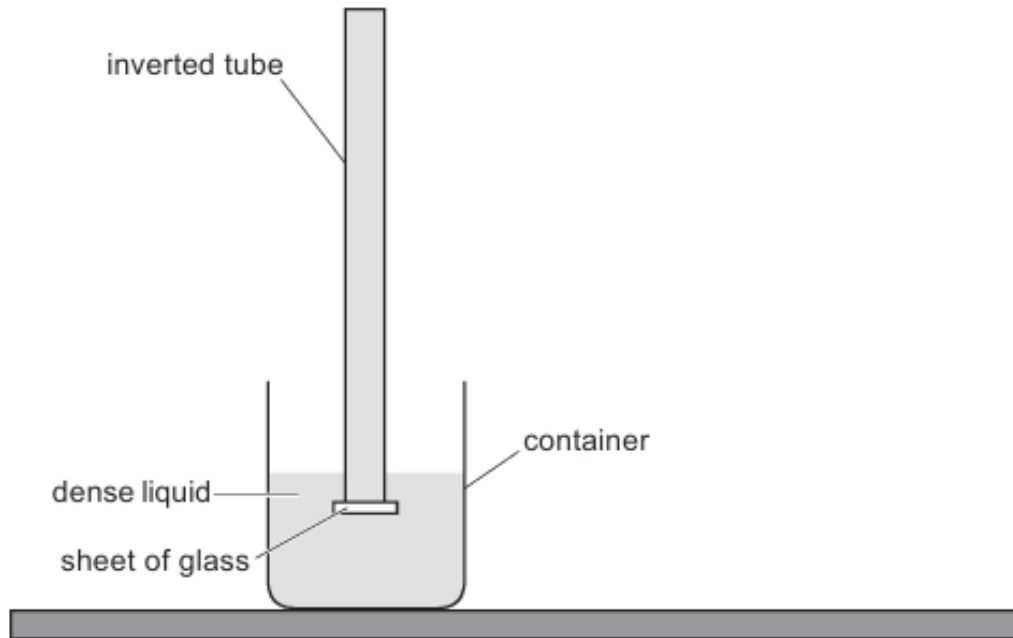


Fig. 4.2 (not to scale)

The sheet of glass seals the open end of the inverted tube.

- (i) Describe what happens in the inverted tube when the sheet of glass is removed.

.....

.....

.....

.....

..... [3]

- (ii) This equipment is used to make a measurement that is used to calculate atmospheric pressure.

Describe the measurement made and then used in the calculation of atmospheric pressure.

You may draw a diagram to help your description.

.....
.....
..... [2]

[Total: 10]

3. M/J/23/21/Q4

- (a) A student pushes a drawing pin into a wooden board, as shown in Fig. 4.1.

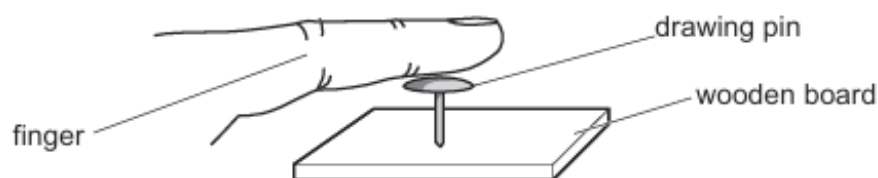


Fig. 4.1

The area of the pin in contact with the finger is $5.0 \times 10^{-5} \text{ m}^2$. The student pushes downwards with a force of 26 N.

The mass of the drawing pin is very small.

- (i) Calculate the pressure exerted by the finger on the drawing pin.

pressure = Pa [2]

- (ii) Compare the force exerted by the finger on the drawing pin with the force exerted by the drawing pin on the wooden board.

.....
..... [1]

- (iii) Explain why the drawing pin goes into the wooden board but not into the finger.

.....
.....
.....
..... [2]

(b) Fig. 4.2 shows water emerging from a plastic bag that contains a number of small holes.

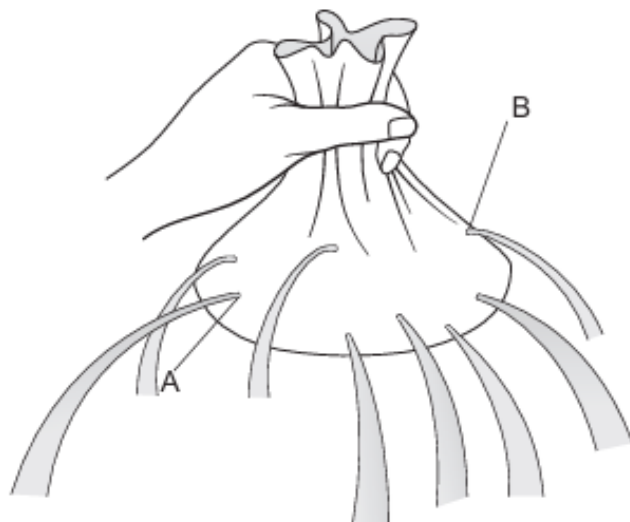


Fig. 4.2

- (i) Explain why the water emerges from each hole in a direction at right angles to the surface of the bag.

.....
.....
.....
..... [2]

- (ii) The holes at A and B are the same size.

Explain why the water emerges faster from the hole at A than from the hole at B.

.....
.....
..... [1]

[Total: 8]

4. M/J/23/22/Q3

A fixed mass of gas in a glass tube is trapped by a seal at one end of the tube and by a column of mercury. The mercury is free to move within the tube.

The tube is rotated slowly from the vertical as shown in Fig. 3.1 to the horizontal as shown in Fig. 3.2. The volume of the gas increases and its temperature remains constant.

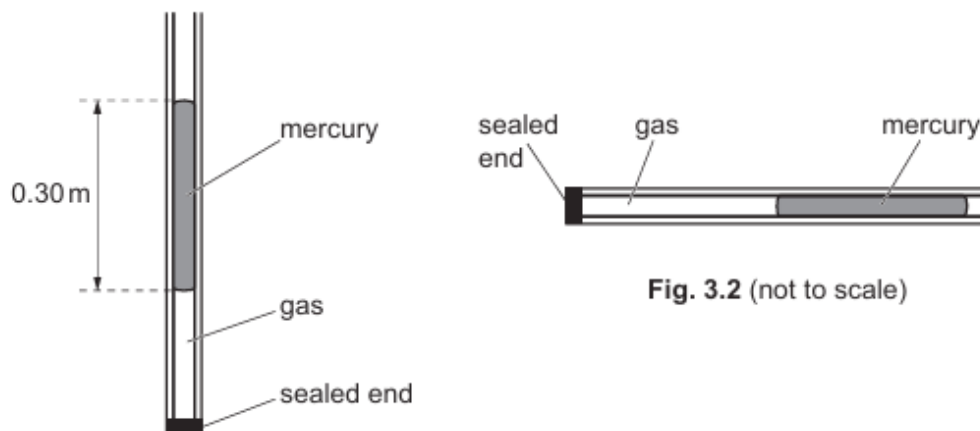


Fig. 3.2 (not to scale)

Fig. 3.1 (not to scale)

- (a) (i) Describe why rotating the tube changes the pressure of the gas in the sealed end.

.....

 [1]

- (ii) Explain, using ideas about particles, why the pressure of the gas decreases when its volume increases.

.....

 [3]

- (b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is $14\,000\text{ kg/m}^3$.

Atmospheric pressure is $1.0 \times 10^5\text{ Pa}$.

Calculate the pressure of the gas in the tube.

pressure = Pa [3]

- (c) The pressure of a different sample of gas changes at constant temperature.

Fig. 3.3 shows one point, marked X, on a graph of pressure against volume for the gas sample.

At X the pressure of the gas is P_0 and its volume is V_0 .

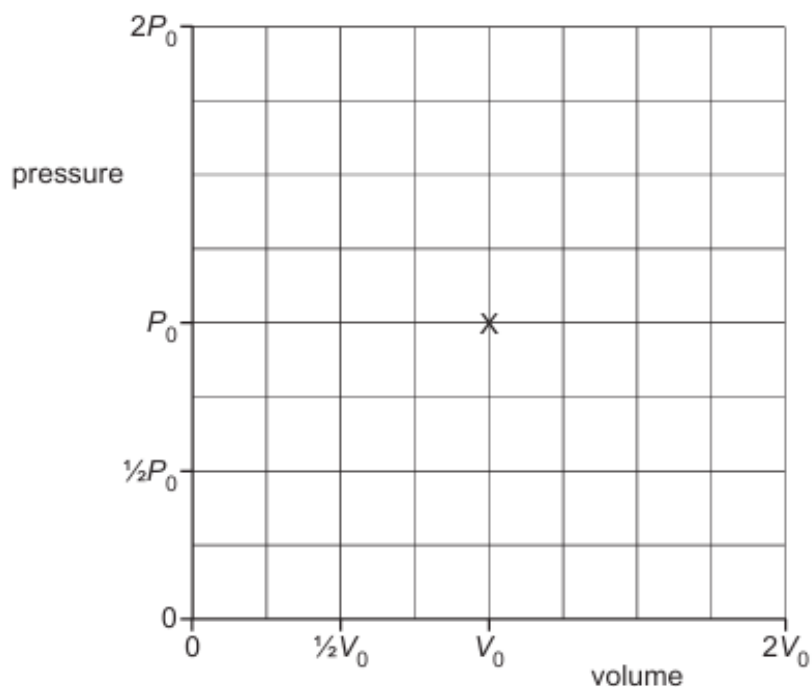


Fig. 3.3

On Fig. 3.3, sketch the graph as the pressure of the gas decreases from $2P_0$ to $\frac{1}{2}P_0$. [2]

[Total: 9]

5. M/J/22/22/Q3

Fig. 3.1 shows a syringe mounted vertically in a block of wood and sealed at one end. A plunger is free to move inside the syringe.

There is trapped air in the syringe.

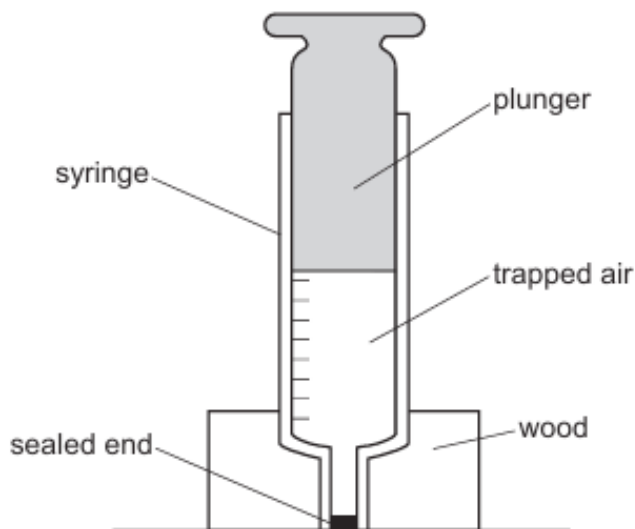


Fig. 3.1

The air inside the syringe exerts a pressure on the walls of the syringe.

(a) Define the term pressure.

.....

.....

..... [1]

(b) Explain how the air molecules in the cylinder of the syringe create a pressure.

.....

.....

.....

.....

..... [3]

- (c) A 10N weight is placed on top of the plunger. The plunger moves down slowly so that the temperature of the air inside the syringe does not change.

Before the weight is placed on top of the plunger:

- the pressure of the air inside the syringe is $1.0 \times 10^5 \text{ Pa}$
- the volume of the air is 50 cm^3 .

The cross-sectional area of the plunger is $1.2 \times 10^{-4} \text{ m}^2$.

- (i) Calculate the pressure of the air in the syringe after the plunger stops moving.

pressure = [2]

- (ii) Calculate the volume of air inside the syringe after the plunger stops moving.

volume = [2]

[Total: 8]

6. O/N/20/21/Q8

The density of water in a lake is 1000 kg/m^3 .

At a depth of 25 m beneath the surface of the lake, the total pressure is $3.5 \times 10^5 \text{ Pa}$.

(a) State what is meant by *pressure*.

.....
..... [1]

(b) The gravitational field strength is 10 N/kg .

Determine:

(i) the pressure due to 25 m of water

pressure = [2]

(ii) the atmospheric pressure.

atmospheric pressure = [1]

(c) An underwater depth gauge contains a small cylinder as shown in Fig. 8.1. Gas is trapped inside the cylinder by a piston. The piston is free to move.

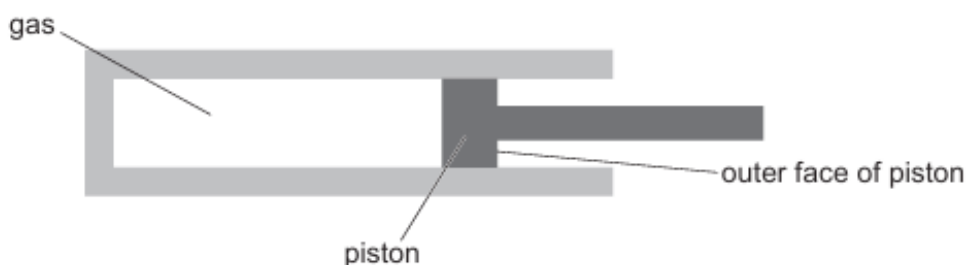


Fig. 8.1

The outer face of the piston is in contact with the water.

As the depth gauge is lowered into the water, the piston moves into the cylinder. This moves a needle on a dial to indicate the depth of the gauge in the water.

- (i) Explain why the piston moves into the cylinder.

.....

.....

.....

.....

..... [3]

- (ii) The temperature of the gas does not change as the piston moves into the cylinder.

Explain, in terms of molecules, what happens to the pressure of the trapped gas as the piston moves into the cylinder.

.....

.....

.....

..... [3]

- (iii) At the surface of the water, the volume of the trapped gas in the depth gauge is V_0 .

On Fig. 8.2, sketch a graph to show how the volume of trapped gas decreases as the gauge is lowered into the water.

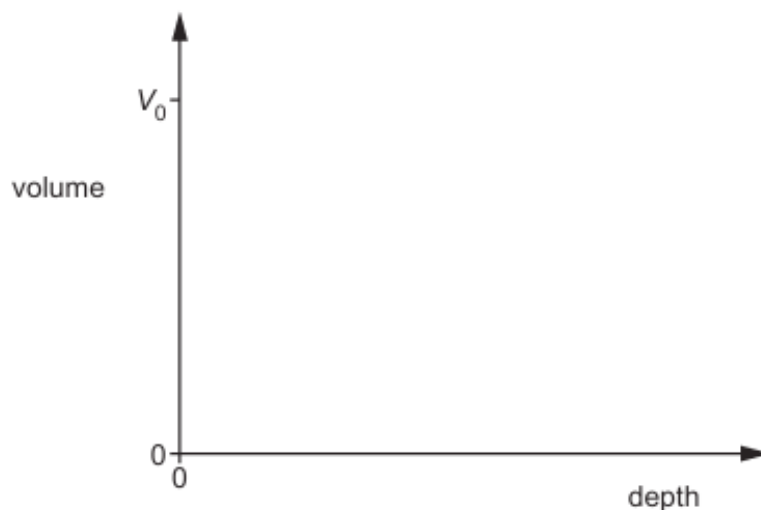


Fig. 8.2

[2]

- (iv) The instructions for the depth gauge state that, each time it is used, the needle of the dial must be re-set to zero at the surface of the water.

Suggest **one** reason for this.

.....

..... [1]

- (v) The density of the air trapped in the depth gauge increases. The density of the water remains constant.

Explain, in terms of the molecules of the water, why the density of the water remains constant.

.....

.....

..... [2]

[Total: 15]

7. O/N/19/21/Q1

Fig. 1.1 shows a tube full of mercury placed upside down in a dish of mercury. Initially a small piece of glass closes the open end of the tube.

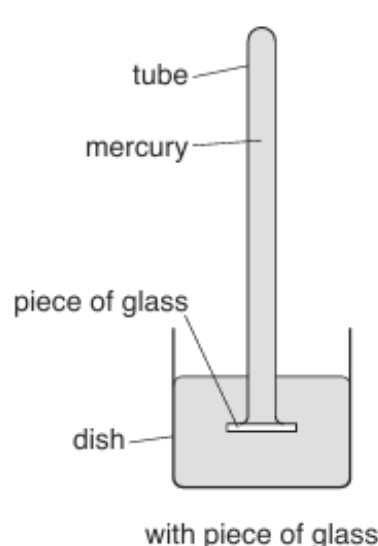


Fig. 1.1

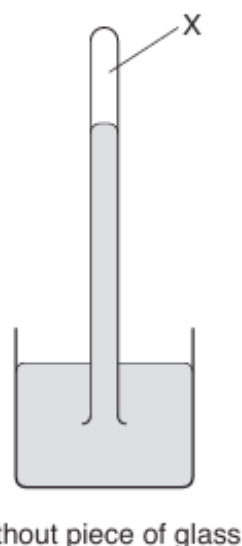


Fig. 1.2

- (a) The piece of glass is removed and the level of mercury in the inverted tube falls, as shown in Fig. 1.2.

- (i) Explain why the level of mercury in the tube falls.

.....

 [2]

- (ii) State what is found at point X in Fig. 1.2.

..... [1]

- (b) Describe how the tube of mercury in Fig. 1.2 is used to determine a value for the atmospheric pressure in pascals (Pa). You may draw on Fig. 1.2 if you wish.

.....

 [3]

[Total: 6]

8. O/N/19/22/Q2

A student uses a pump to inflate a bicycle tyre.

Fig. 2.1 shows the pump and the tyre.

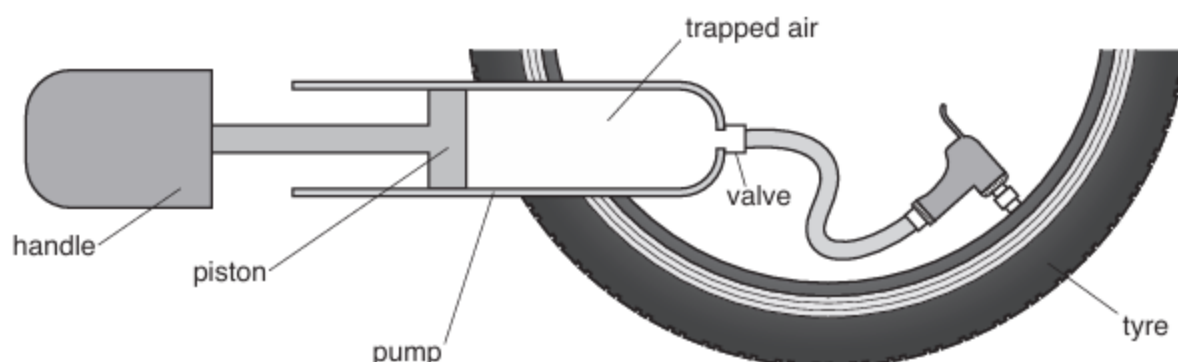


Fig. 2.1 (not to scale)

- (a) The pressure of the trapped air in the pump is $3.8 \times 10^5 \text{ Pa}$ and the cross-sectional area of the piston is $6.1 \times 10^{-4} \text{ m}^2$.

- (i) Calculate the force exerted on the piston by the trapped air in the pump.

force = [2]

- (ii) The student pushes the handle to the right and the piston forces the trapped air into the tyre. The force exerted by the student is less than the value in (a)(i).

Suggest one reason why.

.....
 [1]

- (b) The temperature of the air in the pump remains constant as the handle moves to the right.

Explain, in terms of its molecules, why the pressure of a gas increases when its volume is decreased at constant temperature.

.....

 [3]

[Total: 6]

9. O/N/18/21/Q2

A small crack appears in an oil tank and a stream of oil is pushed out through the crack. The oil hits the floor at P where a puddle of oil starts to form.

Fig. 2.1 shows the oil tank.

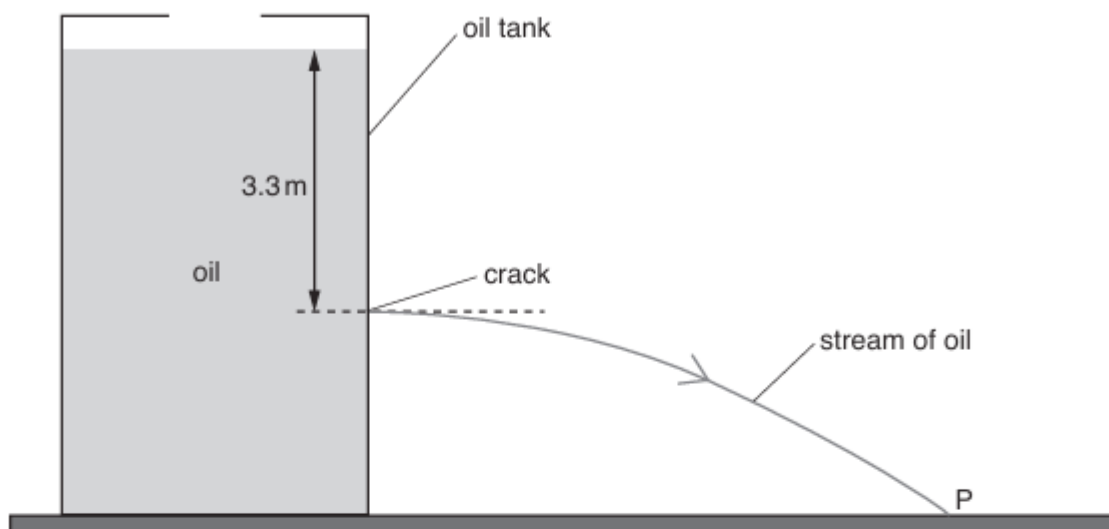


Fig. 2.1

The gravitational field strength g is equal to 10 N/kg .

(a) The density of the oil is 940 kg/m^3 and the crack is 3.3 m below the surface of the oil.

(i) Calculate the pressure due to the oil at the level of the crack.

pressure = [2]

(ii) Explain why the atmospheric pressure does not affect the rate at which the oil is pushed out through the crack.

.....
 [1]

(b) As time passes, the point where the oil hits the floor moves away from P and towards the tank.

Explain why this happens.

.....

 [2]

10. O/N/17/21/Q2

Fig. 2.1 shows a can of compressed air that is being used to blow dust off a computer keyboard.



Fig. 2.1

(a) The pressure of the air inside the can is greater than the pressure of the atmosphere.

(i) State what is meant by the term *pressure*.

.....
..... [1]

(ii) Explain, in terms of molecules, why the pressure of the air inside the can decreases as it is used.

.....
.....
.....
.....
.....
..... [3]

11. O/N/15/22/Q3

A very deep tank is used when training sailors to escape from submarines. The tank is cylindrical and open to the air at the top. Fig. 3.1 shows the tank.

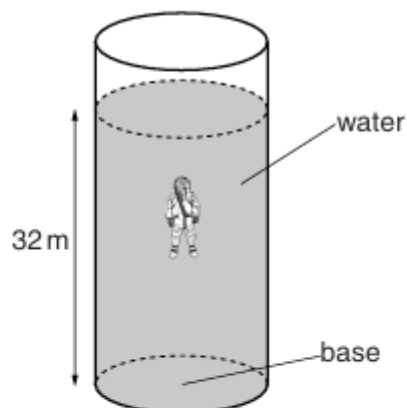


Fig. 3.1 (not to scale)

The area of the base of the tank is 45 m^2 and the tank is filled with water to a depth of 32 m.

- (a)** The density of water is 1000 kg/m^3 and the gravitational field strength g is 10 N/kg .

Calculate the pressure, due to the water, on the base of the tank.

pressure = [2]

- (b)** The pressure in the water at the base of the tank is $4.2 \times 10^5 \text{ Pa}$.

- (i)** Explain why the pressure in the water at the base of the tank differs from the value calculated in **(a)**.

.....
.....
..... [1]

- (ii)** Calculate the force exerted on the base of the tank.

force = [2]

12. M/J/15/22/Q2

Fig. 2.1 shows two engineers measuring the length of a wall made from concrete.

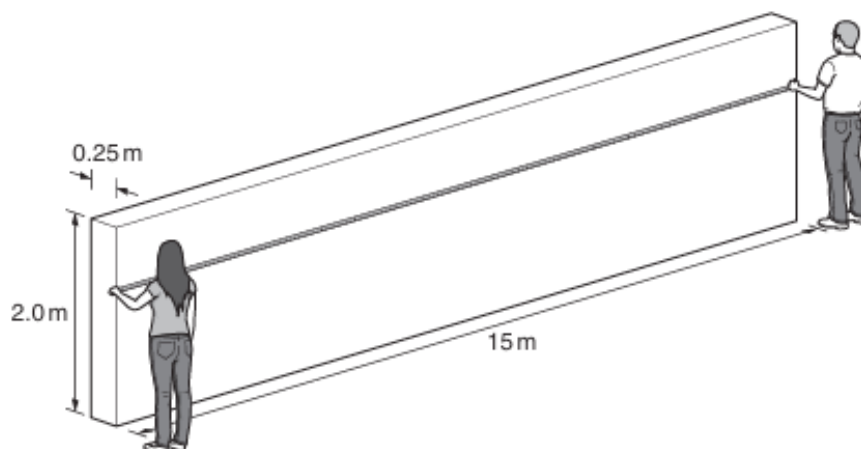


Fig. 2.1 (not to scale)

The wall is 2.0 m high, 15 m long and 0.25 m thick.

The weight of the wall is 180 000 N and the mass of the wall is 18 000 kg.

- (a) The engineers measure the length of the wall in one single measurement.

State the name of the measuring instrument they use.

..... [1]

- (b) The engineers state that the density of the concrete affects the pressure exerted by the wall on the ground but that the length of the wall does not affect this pressure.

- (i) Define *density*.

.....

..... [1]

- (ii) Calculate the density of the concrete.

density = [2]

- (iii) Calculate the pressure exerted by the wall on the ground.

pressure = [2]

- (iv) Without further calculation, explain why doubling the length of the wall does not change the pressure that the wall exerts on the ground.

.....
.....
.....
..... [1]

13. M/J/14/21/Q9

A garden pond contains a small fountain. An electric pump in the water causes the water to rise above the surface of the pond, as shown in Fig. 9.1.

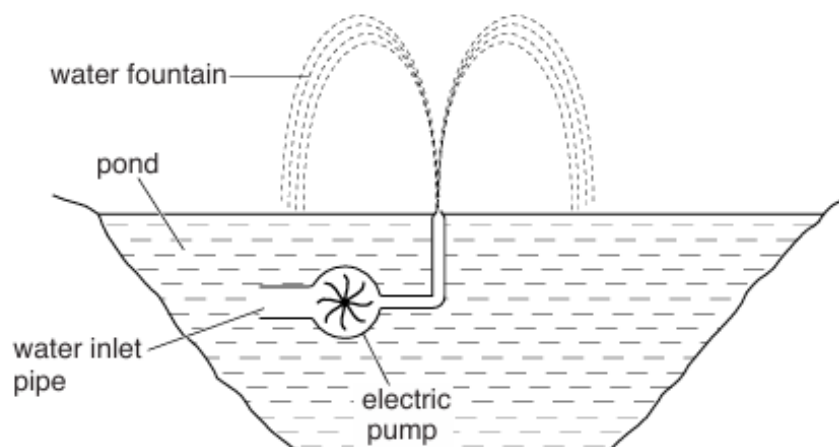


Fig. 9.1

(a) The pressure of the water in the pond increases with depth.

(i) Explain what is meant by *pressure*.

.....
..... [1]

(ii) Explain why the pressure below the surface of the water increases with depth.

.....
..... [2]

(iii) State the unit of pressure.

..... [1]

(b) Describe the energy changes that occur within the pump.

.....
.....
..... [3]

14. M/J/13/21/Q3

Fig. 3.1 shows a glass tube dipped into mercury. A vacuum pump is connected to the top of the tube and switched on. The mercury rises up the tube and stops.

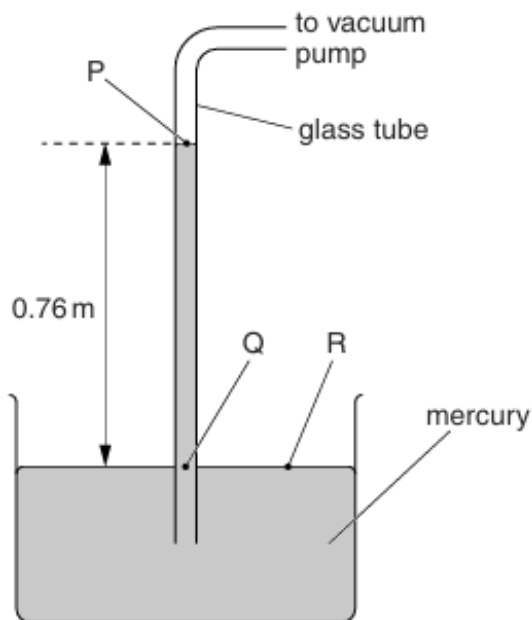


Fig. 3.1

- (a) Three points on Fig. 3.1 are labelled P, Q and R.

State which **two** of these points are at atmospheric pressure.

..... [1]

- (b) The density of mercury is 13600 kg/m^3 and the gravitational field strength g is 10 N/kg .

Calculate the pressure due to the column of mercury of length 0.76 m.

pressure = [2]

- (c) State and explain what happens if the mercury in the apparatus shown in Fig. 3.1 is replaced with water.

.....

 [2]

15. M/J/12/21/Q3

A student measures the pressure inside a bicycle tyre using a pressure gauge he has constructed. Fig. 3.1 shows the apparatus he uses. The piston and rod move along the smooth cylindrical tube.

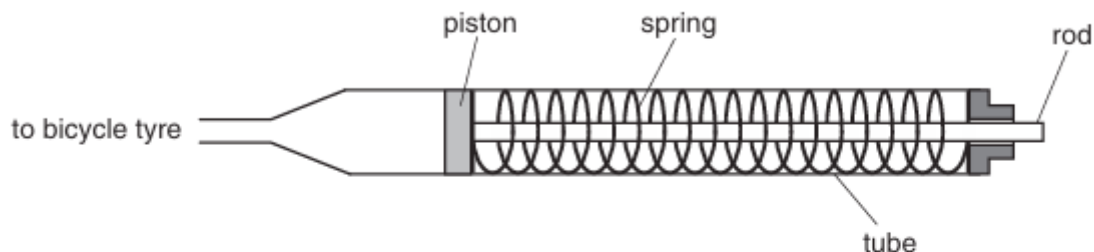


Fig. 3.1

The change in length of the spring is proportional to the force applied to the spring. A force of 2.8 N compresses the spring by 2.0 cm.

The student connects the pressure gauge to the tyre. Air from the tyre exerts a force on the piston. As the piston moves, the spring is compressed. The piston moves 10 cm to the right and then stops.

- (a) Calculate the force exerted on the piston by the spring.

force =[1]

- (b) The cross-sectional area of the piston is $3.0 \times 10^{-5} \text{ m}^2$. Calculate the pressure of the air inside the tyre.

pressure =[2]

- (c) Suggest one change to the apparatus so that the piston moves a smaller distance when measuring the same pressure.

.....[1]

- (d) Each time that the pressure gauge is used, the pressure in the tyre falls slightly. The temperature stays constant. Describe, using ideas about molecules, why the pressure falls.

.....

[2]

16. O/N/09/Q1

A diver holds his breath and dives into the sea from a boat to a depth of 25.0m. The atmospheric pressure is 1.05×10^5 Pa.

- (a) (i) Explain why the pressure at this depth is greater than the atmospheric pressure.

.....
.....[1]

- (ii) Other than the depth and the atmospheric pressure, state **one** quantity that affects the pressure in a liquid.

.....
.....[1]

- (b) (i) The pressure due to 25.0m of sea-water is 2.55×10^5 Pa. Calculate the total pressure at this depth.

pressure =[1]

- (ii) As the diver holds his breath and descends to a depth of 25.0m, the greater pressure causes the volume of the air trapped in his lungs to change.

When he is on the boat, the total volume of the air in his lungs is 6000 cm^3 . Calculate the volume of this air at a depth of 25.0m.

volume =[2]

ANSWERS

1.

Question	Answer	Marks
1(a)(i)	particles move / collide / hit	B1
	particles collide with (inner) walls (of rocket)	B1
	force exerted by collisions over / on an area	B1
1(a)(ii)	more collisions per unit time	B1

2.

Question	Answer	Marks
4(a)(i)	$\Delta p = h\rho g$ or $1.0 \times 1.4 \times 10^4 \times 9.8$	C1
	1.4×10^5 or 1.372×10^5 or $1.0 \times 10^5 + (1.0 \times 1.4 \times 10^4 \times 9.8)$ or $1.0 \times 10^5 + 1.4 \times 10^5$	C1
	2.4×10^5 (Pa)	A1
4(a)(ii)	$(F =) pA$ or $2.4 \times 10^5 \times 4.0 \times 10^{-4}$	C1
	95 (N) or 96 (N)	A1
4(b)(i)	the liquid falls (in the tube)	B1
	then stops falling or reaches stability	B1
	(original) pressure of liquid at bottom of tube greater than pressure of liquid in basin immediately beneath it or pressure (of liquid in tube) becomes equal to atmospheric pressure / pressure in the container or leaving a vacuum (at the closed end of the tube)	B1
4(b)(ii)	height of liquid surface (in tube measured)	B1
	measured relative to level of liquid surface in container or $h\rho g$ (used) to calculate the pressure	B1

3.

Question	Answer	Marks
4(a)(i)	$(P =) \text{ Force / Area}$ algebraic or numerical	C1
	520 000 (Pa)	A1
4(a)(ii)	They are equal (in size and direction)	B1
4(a)(iii)	small(er) area of contact (with board)	C1
	large(r) pressure (on board)	A1
4(b)(i)	<u>pressure</u> (of water) act / <u>pressure</u> in bag	B1
	<u>force</u> (on water) at right angles to the surface / in a straight line or pressure acts in all directions	B1
4(b)(ii)	larger pressure at A / greater weight of water above A (and greater height / depth / mass / weight (of water above B)	B1

4.

Question	Answer	Marks
3(a)(i)	<u>mercury</u> exerts a changing or reduced pressure / force (when vertical but not when horizontal) / <u>mercury</u> stops exerting a pressure	B1
3(a)(ii)	particles hit walls / mercury	B1
	fewer hits per second or hit less often / less frequently or fewer hits per unit area	B1
	less force (per unit area on wall)	B1
3(b)	$(P =) \rho gh$ in any form	C1
	41 000 seen	C1
	140 000 (Pa)	A1
3(c)	curved shape with decreasing gradient	B1
	points plotted at or curve through $(\frac{1}{2}V_0, 2P_0)$, (P_0, V_0) and $(2V_0, \frac{1}{2}P_0)$	B1

5.

Question	Answer	Marks
3(a)	force divided by area	B1
3(b)	(molecules) move (around at random) / have K.E.	B1
	(molecules) hit sides / piston	B1
	(molecules) cause force on piston / sides	B1
3(c)(i)	$10 / 1.2 \times 10^{-4}$ or 8.3×10^4 (Pa) seen	C1
	1.8×10^5 Pa	A1
3(c)(ii)	$P_1 V_1 = P_2 V_2$ algebraic or numerical in any form or $P \propto 1 / V$	B1
	27 or 28 cm ³	B1

6.

Question	Answer	Marks
8(a)	(normal) force per unit area or (normal) force $\div$ area	B1
8(b)(i)	$(p =) \rho gh$ or $25 \times 1000 \times 10$	C1
	2.5×10^5 Pa	A1
8(b)(ii)	1.0×10^5 Pa	B1
8(c)(i)	pressure increases with depth	B1
	pressure / force on outer face greater than that on inner face	B1
	resultant force on piston	B1
8(c)(ii)	pressure increases and more collisions (of molecules with walls of cylinder)	B1
	more frequent collisions or more collisions per unit area	B1
	molecules have less distance to travel between collisions (with walls) or molecular density greater	B1
8(c)(iii)	<u>curve</u> with negative gradient and gradient of decreasing magnitude	B1
	<u>negative gradient</u> at V_0 and approaches x-axis asymptotically	B1
8(c)(iv)	atmospheric pressure varies or temperature varies	B1
8(c)(v)	molecules touching or (very) close together (and cannot be forced closer)	B1
	(repulsive) forces between molecules (very) large	B1

7.

Question	Answer	Marks
1(a)(i)	pressure due to mercury (in tube) is greater than pressure due to atmosphere (+ small height of mercury)	B1
	(resultant) force downwards and mercury flows out of tube	B1
1(a)(ii)	vacuum or nothing or mercury vapour	B1
1(b)	height of mercury column mentioned	C1
	height of mercury column <u>above level in dish measured</u>	A1
	candidate suggests use of $p = h \rho g$ and h in metres (m)	B1

8.

Question	Answer	Marks
2(a)(i)	$(F =) pA$ or $3.8 \times 10^5 \times 6.1 \times 10^{-4}$	C1
	230 N	A1
2(a)(ii)	force on other side of piston due to air / atmospheric pressure or pressure of trapped air decreases as valve opens (and air enters the tyre)	B1
2(b)	molecules collide <u>with walls / piston / pump / surface</u>	B1
	number of molecules per unit volume increases (as volume decreases)	B1
	more collisions per unit area or more frequent collisions / collide more often	B1

9.

Question	Answer	Marks
2(a)(i)	$(p =) \rho gh$ or $940 \times 10 \times 3.3$	C1
	3.1×10^4 Pa	A1
2(a)(ii)	atmospheric pressure acts at both ends of crack	B1
2(b)	(surface) level decreases / drops	B1
	pressure (at level of crack) decreases	B1

10.

Question	Answer	Marks
2(a)(i)	(pressure =)force / area	B1
2(a)(ii)	fewer molecules or less gas or more space / further apart	B1
	less frequent collisions (with walls)	B1
	less force exerted on walls	B1

11.

(a)	$(p =)h\rho g$ or $32 \times 1000 \times 10$	C1
	3.2×10^5 Pa	A1
(b) (i)	atmospheric pressure (is also acting on the surface of the water)	B1
(ii)	$(F =)pA$ or $4.2 \times 10^5 \times 45$ or $3.2 \times 10^5 \times 45$ or 1.44×10^7	C1
	$1.9 / 1.89 \times 10^7$ N	A1

- 12.**
- (a) tape B1
- (b) (i) mass $\div$ volume **or** mass per unit volume B1
- (ii) (V=) $15 \times 0.25 \times 2$ or 7.5 seen C1
 2400 kg/m^3 A1
- (iii) (A=) 15×0.25 **or** 3.75 **or** (P=) F/A C1
or (P=) $180000/A$ **or** (P=) $dgh / \rho gh$ seen A1
48000 Pa
- (iv) (length doubles) so both area **and** weight/force double B1
or area **and** force/weight both increase/larger (in proportion)
or height **and** density the same (in $P = dgh$)
- 13.**
- (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]
- (ii) more water above **or** more force from water **or** more atoms/molecules above [C1]
larger weight of water above [A1]
- (iii) Pa **or** N/m^2 [B1]
- 14.**
- (a) Q and R B1
- (b) ρgh in any form, algebraic or numerical C1
 $1.0(336) \times 10^5 \text{ N/m}^2$ A1
- (c) water is less dense **or** has density $1000 \text{ (kg/m}^3\text{)}$ B1
water further up tube/fills tube **or** height greater **or** water enters pump **or** water boils B1

- 15.**
- (a) 14 N B1
- (b) ($P =$) F/A algebraically in symbols or words in any form; $14/3.0 \times 10^{-5}$ C1
 $4.67 \times 10^5 \text{ Pa}$; $4.7 \times 10^5 \text{ Pa}$ ecf (a) A1
- (c) stiffer/stronger spring; piston has less area/diameter; smaller piston (and tube) B1
- (d) molecules/particles/atoms collide with tyre/walls/piston B1
 air/molecules enter gauge; fewer molecules in the tyre; fewer hits/sec;
 less frequent hits; volume increases B1
- 16.**
- (a) (i) weight of water (causes extra pressure)(**not** mass) B1
- (ii) density of liquid/(sea-)water **or** gravitational field strength/acceleration of freefall
 (**not** gravity) B1
- (b) (i) $3.6/3.60 \times 10^5 \text{ Pa}$ B1
- (ii) $P_1 V_1 = P_2 V_2$ **or** $1.05 \times 10^5 \times 6000 = 3.60 \times 10^5 \times V_2$ C1
 1700 **or** 1750 **or** 1800 cm^3 A1